Operation and Maintenance Manual

Translation of the Original Operating Manual

IITJMU/CE/SP-17/2020-21

Low Velocity Impact Testing Machine

Indigenous Innovative Technology Methodology

Newtomech

"Best solutions for your success"

Newtomech Innovative Solutions Private Limited

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INDIAN INSTITUTE OF TECHNOLOGY MADRAS

RTBI

IITM's Rural Technology and Business Incubator

iitm research park

Bringing unlike minds together

Products

1. Dynamic Testing

- Drop mass set-up for low velocity impact testing
- Gas gun for medium and high velocity impact testing
- Hopkinson bar test set-up for high strain rate tensile and compression testing

2. Societal impact products

- Hk-NoT – A safety cut-off device for Ceiling Fans (About 10000 devices have been installed in leading academic Institutions)
- Other products are under R&D

3. Industrial support

- Design Evaluation for Certification of products and equipment
- Consultancy for Indigenous product development

3. Mechanical Technical Specifications:

S. No	Description	Specification
1	Drop Height	2400mm
2	Drop Height Accuracy	± 1 mm
3	Drop Height Resolution	1 mm
4	Drop Mass	50 kg (Max) with the step of 5 kg for polymer composite (PMC) specimens and 350 kg (Max) with the step of 50 kg for concrete specimens.
5	Maximum Drop Velocity	6.5-6.85 m/s
6	Maximum Drop Energy	1000 -1200 J for PMC specimen and 8000-8240 J for Concrete specimen.
7	Stroke (adjustable)	300 mm -2400 mm
8	Striker/Tup diameter	Hemispherical 12.7mm dia, 60-62 HRC for PMC and 50 mm thickness and 150 mm width with 25 mm radius at contact point of Concrete specimens, 60-62 HRC.
9	Tup weight	5.5 kg ± 0.25 kg For PMC, and 55 kg ± 0.5 kg for Concrete specimens.
10	Linear Displacement Sensor (Wenglor)	Non-contact Type
11	Accelerometer Maximum range	10000 g
12	DAQ system to generate graphs (Load Vs Time, Acceleration Vs time, Load vs. Displacement etc.)	1
13	Compatible- signal Conditioner for Force Sensor/inbuilt	1
14	Compatible- signal Conditioner for displacement/inbuilt	1
15	Compatible- signal Conditioner for displacement/inbuilt	1
16	Compatible- Signal Conditioner for acceleration/inbuilt	1
17	PC to store LabVIEW data and store the generated curves. Official latest Windows OS, i7 or equivalent processor, 16 GB RAM and 500 GB Hard drive. Monitor 15", keyboard, mouse and UPS.	1
18	Mechanical frame System to withstand impact of 350 kg drop mass (four columns)	1 set
19	A test fixture setup to place concrete specimens of size 150 x 150 x 1100 mm.	1 set
20	Three point bending fixture for polymer matrix test (PMC) specimens of size 210x60x10mm.	1set
21	Impact testing fixture for polymer matrix to test specimens of size as per ASTMD7136/D7136M-15.	1 set

4. Safety Precautions

Definition of precautionary information

The following notation is used in this manual to provide precautions required to ensure safe usage of Drop Tower Operator and the Drop Tower Controller.

The safety precautions that are provided are extremely important to ensure safety. Always read andheed the information provided in all safety precautions.

The following notation is used.

WARNING	Indicates a potentially hazardous situation which, if not avoided, could result in death or serious injury. Additionally, there may besevere property damage.
Caution	Indicates a potentially hazardous situation which, if not avoided, may result in minor or moderate injury, or property damage.

Precautions for Safe Use

Indicates precautions on what to do and what not to do to ensure safe usage of the product.

Precautions for Correct Use

Indicates precautions on what to do and what not to do to ensure proper operation and performance.

5. Symbols

The circle and slash symbol indicates operations that you must not do.The specific operation is shown in the circle and explained in text.
This example indicates prohibiting disassembly.

This example indicates a precaution against electric shock. The specific operation is shown in the triangle.

The triangle symbol indicates precautions (including warnings).

The filled circle symbol indicates operations that you must do. The specific operation is shown in the circle and explained in text.
This example shows a general precaution for something that you must do.

⚠ WARNING

Check the NC switches for proper execution before you use it for actual operation.

The devices or machines may perform unexpected operations regardless of the operating mode of the CPU Unit.

Always confirm safety before you reset the system.

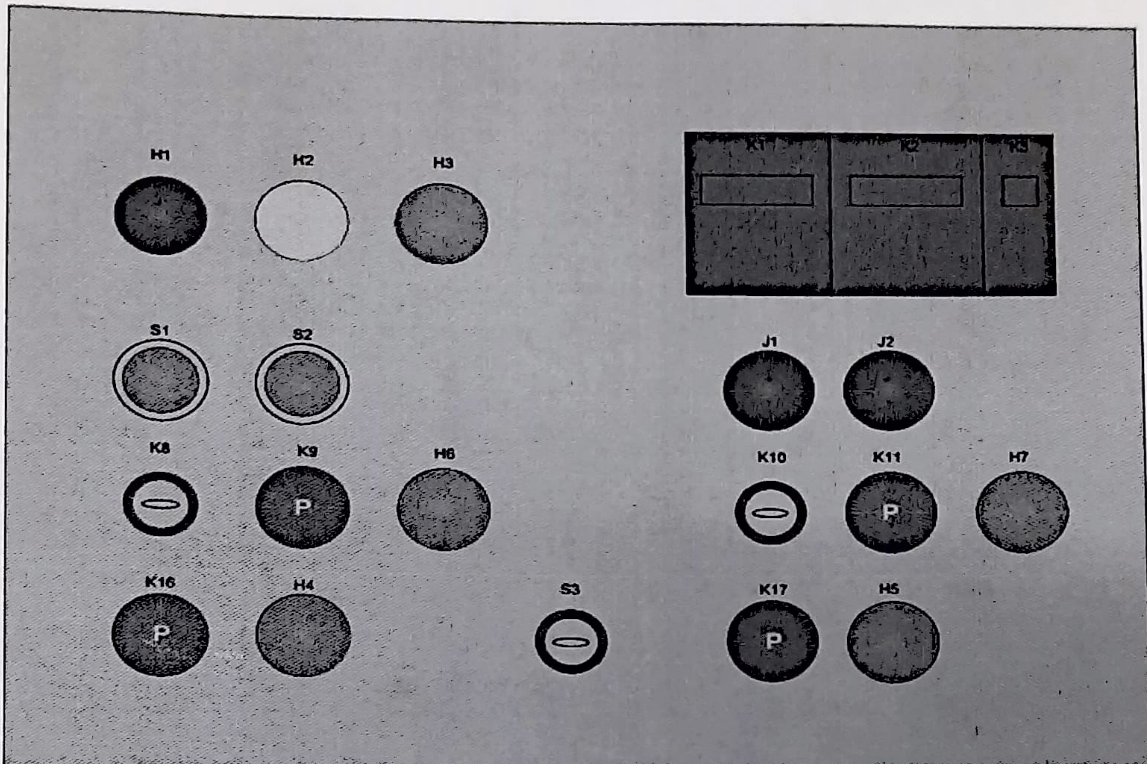
Always confirm safety at the destination node before you operate the drop tower.

6. Electrical Technical Specification:

S.No	Description	HSN Code	Denoted As
1	LED Indicator Red	85361090	H1
2	LED Indicator Yellow	85361090	H2
3	LED Indicator Green	85361090	H3
4	MCB 16A 3Pole	85362329	K1
5	MCB 16A 3Pole	85362329	K2
6	MCB 16A 1Pole	85362329	K3
7	Push button switch with indicator	85361090	S1
8	Siemens contactor 3TF	85365010	K4 & K6
9	Siemens contactor 3TF	85365010	K5 & K7
10	Push button switch with indicator	85361090	S2
11	Compressor Motor		M1
12	Limit switch 8108	85365090	J1
13	Siemens contactor 3TF	85365010	K12 & K14
14	Siemens contactor 3TF	85365010	K13 & K15
15	Push button switch	82361090	K16
16	Push button switch	82361090	K17
17	LED Indicator Green	85361090	H9
18	LED Indicator Red	85361090	H4
19	Hoist motor		M2
20	Limit switch 8108	85365090	J2
21	LED Indicator Green	85361090	H8
22	LED Indicator Red	85361090	H5
23	SMPS 2A 24V	85381090	G1
24	SPST Toggle switch	85369090	K8
25	SPST Toggle switch	85369090	K10
26	push button switch	82361090	K9
27	push button switch	82361090	K11
28	LED Indicator Green	85361090	H6
29	LED Indicator Green	85361090	H7
30	DPDT Toggle Switch		S3
31	Air reservoir		T1
32	1/2" FLR unit Techno	84799090	F1
33	1/4" 2 Station Manifold SMC	84811000	Q1
34	1/4" 5/2 Single Solenoid Valve 24V DC SMC	84799090	V1
35	1/4" 5/2 Single Solenoid Valve 24V DC SMC	84799090	V2
36	Festo cylinder		C1
37	Festo cylinder		C2

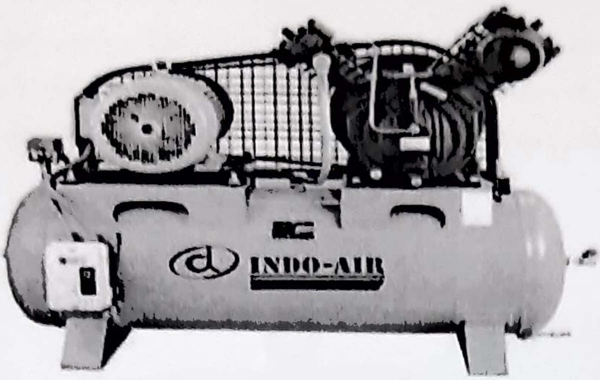
7. Control Panel Information

The control panel is made with high quality metal and it is fixed with indicator and controlling switches to operate the drop tower. The operating details and panel figure are given below. An key will be provide with the panel, which will be used to open the panel and it should only with the authority of drop tower.



- H1, H2, & H3 are used to indicate the R, Y, & B terminals which is connected to the main supply.
- K1, K2, & K3 are the tripper switches which will switch ON the power path of Air compressor, Hoist motor & Pneumatic control respectively.
- S1 is the push button switch with indicator, which energize the contactor and closes the path of Air compressor motor.
- S2 is the push button switch with indicator, which energize the contactor and closes the path of Hoist motor.
- J1 & J2 are the limit switch indicators for upper limit and bottom limit respectively.
- K8 & K9 are the SPST switches which is used to trigger the Unlock switches, and it is represented with indicator H6 & H7. K9 & K11 are the push button Unlock switches.
- The SPDT switch S3 is used to select the forward and reverse operation of Hoist motor.
- K16 & K17 are operator switches of forward and reverse direction respectively with the indicators H4 and H5.

10. Compressor

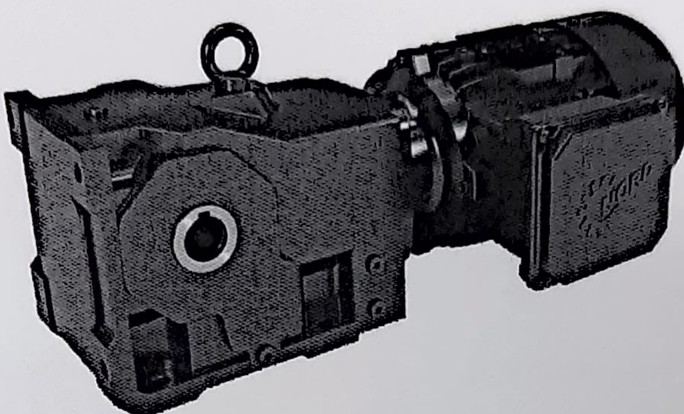


MODEL	MOTOR		NO. OF	RPM	RECIEVER		MAX PRESSURE		PISTON DISPL		
	HP	KW	CYL.		CFT.	LTRS.	PSIG	KG/CM^2	CFM	M^3/M	LTR/MIN
IA 10 NL	3	2.2	2	660	7.95	225	142	10	13.20	0.374	374

- Sealed and high temperature resistant pre-lubricated bearings.
- Well-designed cooling system of air cooled inter/after coolers.
- Special material piston ring which retains its properties under high temperature working conditions.
- Piston ring outer surface always in contact with the cylinder bore even after thousands of hours of running and corresponding wear ensuring unchanged pumping up time.
- Special material piston ensures the retention of cylinder bore finish in adverse temperature condition and expansion of piston.

10. Hoist Motor

NORD drive solutions are of the highest precision and are optimally suited for use in demanding applications. NORD drive components are calculated according to DIN 3990, NIEMANN and AGMA, and the certification according to DIN ISO 9001 documents that have applied the quality management consistently. Only high-quality materials and standard parts are used within production – after careful checking of quality, structure and rigidity – and all production stages are subjected to hard quality controls. You can rely on thoroughly checked and practical functionality and operational reliability.



MODEL	MOTOR		R/MIN	RECIEVER	NM
	HP	KW		R/MIN	
SK80LH/4	1	0.75	1415	32	167

12. Safety features

- If any overload or short circuits occurs the tripper will cut-off from mains supply.
- Separate switches and contactors are used.
- Forward or Reverse, only anyone operation can be done at a time for motor safety.
- The hoist will always be locked with load until the trigger is not done and also will be locked even if any sudden power disturbance occurs.

13. Repair information

For enquiries to our technical and mechanical service departments, please have the order number to hand.

Repairs

The device must be sent to the following address if it needs repairing:

Newtomech Innovative Solutions Private Limited
IIT Madras Research Park,
Kanagam, Taramani, Chennai-600 113, India.
+91-9940032476/9839844245

ⓘ Information

No guarantee can be given for any attachments, such as encoders or external devices, if a gear unit orgeared motor is sent for repair.

This is important in order to keep repair times as short and efficient as possible.

14. Warranty

Newtomech Innovative Solutions Private Limited accepts no liability for damage to persons, materials or assets as a result of failure to observe this operating manual, operating errors or incorrect use. General wearing parts, e.g. radial seals are excluded from the warranty.